

**Supporting Information for the publication entitled:
Methane Activation at the Pd/CeO₂ Interface**

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This file contains the ReaxFF force-field file for Pd/O/C/H

Reactive MD-force field: Pd/O/C/H

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39      ! Number of general parameters
50.0000 !Overcoordination parameter
 9.5469 !Overcoordination parameter
26.5405 !Valency angle conjugation parameter
 1.7224 !Triple bond stabilisation parameter
 6.8702 !Triple bond stabilisation parameter
60.4850 !C2-correction
 1.0588 !Undercoordination parameter
 4.6000 !Triple bond stabilisation parameter
12.1176 !Undercoordination parameter
13.3056 !Undercoordination parameter
-70.5044 !Triple bond stabilization energy
 0.0000 !Lower Taper-radius
10.0000 !Upper Taper-radius
 2.8793 !Not used
33.8667 !Valency undercoordination
 6.0891 !Valency angle/lone pair parameter
 1.0563 !Valency angle
 2.0384 !Valency angle parameter
 6.1431 !Not used
 6.9290 !Double bond/angle parameter
 0.3989 !Double bond/angle parameter: overcoord
 3.9954 !Double bond/angle parameter: overcoord
-2.4837 !Not used
 5.7796 !Torsion/BO parameter
10.0000 !Torsion overcoordination
 1.9487 !Torsion overcoordination
-1.2327 !Conjugation 0 (not used)
 2.1645 !Conjugation
 1.5591 !vdWaals shielding
 0.1000 !Cutoff for bond order (*100)
 2.1365 !Valency angle conjugation parameter
 0.6991 !Overcoordination parameter
50.0000 !Overcoordination parameter
 1.8512 !Valency/lone pair parameter
 0.5000 !Not used
20.0000 !Not used
 5.0000 !Molecular energy (not used)
 0.0000 !Molecular energy (not used)
 2.6962 !Valency angle conjugation parameter
5      ! Nr of atoms; cov.r; valency;a.m;Rvdw;Evdw;gammaEEM;cov.r2;#
      alfa;gammavdW;valency;Eunder;Eover;chiEEM;etaEEM;n.u.
      cov r3;Elp;Heat inc.;n.u.;n.u.;n.u.;n.u.
      ov/un;vall;n.u.;val3,vval4
C      1.3817   4.0000  12.0000   1.8903   0.1838   0.9000   1.1341
4.0000
      9.7559   2.1346   4.0000  34.9350  79.5548   5.9666   7.0000
0.0000
      1.2114   0.0000 202.5551   8.9539  34.9289  13.5366   0.8563
0.0000
      -2.8983   2.5000   1.0564   4.0000   2.9663   0.0000   0.0000
0.0000
```

H	0.8930	1.0000	1.0080	1.3550	0.0930	0.8203	-0.1000
1.0000							
	8.2230	33.2894	1.0000	0.0000	121.1250	3.7248	9.6093
1.0000							
	-0.1000	0.0000	61.6606	3.0408	2.4197	0.0003	1.0698
0.0000							
	-19.4571	4.2733	1.0338	1.0000	2.8793	0.0000	0.0000
0.0000							
O	1.2450	2.0000	15.9990	2.3890	0.1000	1.0898	1.0548
6.0000							
	9.7300	13.8449	4.0000	37.5000	116.0768	8.5000	8.3122
2.0000							
	0.9049	0.4056	59.0626	3.5027	0.7640	0.0021	0.9745
0.0000							
	-3.5500	2.9000	1.0493	4.0000	2.9225	0.0000	0.0000
0.0000							
X	-0.1000	2.0000	1.0080	2.0000	0.0000	1.0000	-0.1000
6.0000							
	10.0000	2.5000	4.0000	0.0000	0.0000	3.5000	85.0000
0.0000							
	-0.1000	0.0000	-2.3700	8.7410	13.3640	0.6690	0.9745
0.0000							
	-11.0000	2.7466	1.0338	6.2998	2.8793	0.0000	0.0000
0.0000							
Pd	1.8582	2.0000	106.4200	2.0113	0.2465	1.0000	-1.0000
2.0000							
	12.5712	6.0083	2.0000	0.0000	0.0000	5.5005	6.6477
0.0000							
	-1.0000	0.0000	89.8000	27.2704	1.8727	0.1586	0.8563
0.0000							
	-13.0000	1.7287	1.0338	5.0000	2.5791	0.0000	0.0000
0.0000							
10	! Nr of bonds; Edis1;LPpen;n.u.;pbe1;pbo5;13corr;pbo6						
	pbe2;pbo3;pbo4;n.u.;pbo1;pbo2;ovcorr						
1 1	158.2004	99.1897	78.0000	-0.7738	-0.4550	1.0000	37.6117
0.4147							
	0.4590	-0.1000	9.1628	1.0000	-0.0777	6.7268	1.0000
0.0000							
1 2	169.4760	0.0000	0.0000	-0.6083	0.0000	1.0000	6.0000
0.7652							
	5.2290	1.0000	0.0000	1.0000	-0.0500	6.9136	0.0000
0.0000							
2 2	153.3934	0.0000	0.0000	-0.4600	0.0000	1.0000	6.0000
0.7300							
	6.2500	1.0000	0.0000	1.0000	-0.0790	6.0552	0.0000
0.0000							
1 3	158.6946	107.4583	23.3136	-0.4240	-0.1743	1.0000	10.8209
1.0000							
	0.5322	-0.3113	7.0000	1.0000	-0.1447	5.2450	0.0000
0.0000							
3 3	142.2858	145.0000	50.8293	0.2506	-0.1000	1.0000	29.7503
0.6051							
	0.3451	-0.1055	9.0000	1.0000	-0.1225	5.5000	1.0000
0.0000							

2	3	160.0000	0.0000	0.0000	-0.5725	0.0000	1.0000	6.0000
0.5626		1.1150	1.0000	0.0000	0.0000	-0.0920	4.2790	0.0000
0.0000								
1	5	195.8048	0.0000	0.0000	-0.2150	-0.2000	1.0000	16.0000
0.1001		10.7579	-0.2000	15.0000	1.0000	-0.1606	4.0000	1.0000
0.0000								
2	5	67.5406	0.0000	0.0000	0.6327	0.0000	1.0000	6.0000
0.2077		0.7493	1.0000	0.0000	1.0000	-0.0547	4.0000	0.0000
0.0000								
3	5	56.7412	0.0000	0.0000	0.7149	-0.2000	1.0000	16.0000
0.1000		11.9839	-0.2000	15.0000	1.0000	-0.3000	4.7089	1.0000
0.0000								
5	5	90.7003	0.0000	0.0000	-0.1661	-0.2000	0.0000	16.0000
0.2578		3.0618	-0.2000	15.0000	1.0000	-0.0914	5.4665	0.0000
0.0000								
6		! Nr of off-diagonal terms; Ediss;Ro;gamma;rsigma;rpi;rpi2						
1	2	0.1239	1.4004	9.8467	1.1210	-1.0000	-1.0000	
1	3	0.1156	1.8520	9.8317	1.2854	1.1352	1.0706	
2	3	0.0283	1.2885	10.9190	0.9215	-1.0000	-1.0000	
1	5	0.1169	2.3000	9.2890	1.6264	-1.0000	-1.0000	
2	5	0.1445	1.1103	13.0000	1.5714	-1.0000	-1.0000	
3	5	0.2237	1.6610	10.6568	1.7139	-1.0000	-1.0000	
36		! Nr of angles;at1;at2;at3;Thetao,o;ka;kb;pv1;pv2						
1	1	1	59.0573	30.7029	0.7606	0.0000	0.7180	6.2933
	1	2	65.7758	14.5234	6.2481	0.0000	0.5665	0.0000
	2	1	70.2607	25.2202	3.7312	0.0000	0.0050	0.0000
	1	2	0.0000	0.0000	6.0000	0.0000	0.0000	0.0000
	1	2	0.0000	3.4110	7.7350	0.0000	0.0000	0.0000
	2	2	0.0000	27.9213	5.8635	0.0000	0.0000	0.0000
	1	1	49.6811	7.1713	4.3889	0.0000	0.7171	10.2661
	3	1	77.7473	40.1718	2.9802	-25.3063	1.6170	-46.1315
	2	1	65.0000	13.8815	5.0583	0.0000	0.4985	0.0000
	1	3	73.5312	44.7275	0.7354	0.0000	3.0000	0.0000
	1	3	79.4761	36.3701	1.8943	0.0000	0.7351	67.6777
	3	3	80.7324	30.4554	0.9953	0.0000	1.6310	50.0000
	1	3	70.1880	20.9562	0.3864	0.0000	0.0050	0.0000
	2	3	75.6935	50.0000	2.0000	0.0000	1.0000	0.0000
	2	3	85.8000	9.8453	2.2720	0.0000	2.8635	0.0000
	1	2	0.0000	25.0000	3.0000	0.0000	1.0000	0.0000
	3	2	0.0000	15.0000	2.8900	0.0000	0.0000	0.0000
	2	2	0.0000	8.5744	3.0000	0.0000	0.0000	0.0000
	5	3	53.8540	30.0000	2.0649	0.0000	1.0000	0.0000
	3	5	55.7429	4.8907	3.9242	0.0000	0.1000	0.0000
	3	3	90.0000	30.0000	1.6813	0.0000	1.0000	0.0000
	3	5	0.0100	30.0000	8.0000	0.0000	1.0000	0.0000
	2	5	25.6680	10.1984	7.6844	0.0000	1.0000	0.0000
	2	2	83.9156	13.0474	0.9983	0.0000	1.0001	0.0000
	5	2	0.9367	12.0943	6.6344	0.0000	1.0000	0.0000
	2	5	40.0354	9.2882	8.0000	0.0000	1.0000	0.0000

1	1	5	23.3470	30.0000	4.9485	0.0000	0.1000	0.0000	2.6345
3	1	5	0.0100	30.0000	7.8334	0.0000	0.1000	0.0000	1.1134
1	5	5	81.4675	17.7241	8.0000	0.0000	0.1000	0.0000	3.3824
1	5	1	76.1959	30.0000	8.0000	0.0000	0.1000	0.0000	1.3967
5	1	5	0.0100	30.0000	7.8334	0.0000	0.1000	0.0000	1.1134
2	1	5	81.2014	30.0000	0.5384	0.0000	1.0000	0.0000	1.9241
1	5	2	90.0000	0.2503	0.0100	0.0000	1.0000	0.0000	2.1438
1	2	5	6.0617	0.0100	7.5704	0.0000	0.0000	0.0000	3.9408
2	3	5	50.4877	0.1082	1.9877	0.0000	1.0000	0.0000	1.3192
2	5	3	17.8836	0.0100	1.7426	0.5000	1.0000	0.0000	1.1872
25	! Nr of torsions;at1;at2;at3;at4;;V1;V2;V3;V2 (BO);vconj;n.u;n								
1	1	1	1	-0.2500	34.7453	0.0288	-6.3507	-1.6000	0.0000
0.0000									
1	1	1	2	-0.2500	29.2131	0.2945	-4.9581	-2.1802	0.0000
0.0000									
2	1	1	2	-0.2500	31.2081	0.4539	-4.8923	-2.2677	0.0000
0.0000									
1	1	1	3	-0.3495	22.2142	-0.2959	-2.5000	-1.9066	0.0000
0.0000									
2	1	1	3	0.0646	24.3195	0.6259	-3.9603	-1.0000	0.0000
0.0000									
3	1	1	3	-0.5456	5.5756	0.8433	-5.1924	-1.0180	0.0000
0.0000									
1	1	3	1	1.7555	27.9267	0.0072	-2.6533	-1.0000	0.0000
0.0000									
1	1	3	2	-1.4358	36.7830	-1.0000	-8.1821	-1.0000	0.0000
0.0000									
2	1	3	1	-1.3959	34.5053	0.7200	-2.5714	-2.1641	0.0000
0.0000									
2	1	3	2	-2.5000	70.0597	1.0000	-3.5539	-2.9929	0.0000
0.0000									
1	1	3	3	0.6852	11.2819	-0.4784	-2.5000	-2.1085	0.0000
0.0000									
2	1	3	3	0.1933	80.0000	1.0000	-4.0590	-3.0000	0.0000
0.0000									
3	1	3	1	-1.9889	76.4820	-0.1796	-3.8301	-3.0000	0.0000
0.0000									
3	1	3	2	0.2160	72.7707	-0.7087	-4.2100	-3.0000	0.0000
0.0000									
3	1	3	3	-2.5000	71.0772	0.2542	-3.1631	-3.0000	0.0000
0.0000									
1	3	3	1	2.5000	-0.6002	1.0000	-3.4297	-2.8858	0.0000
0.0000									
1	3	3	2	-2.5000	-3.3822	0.7004	-5.4467	-2.9586	0.0000
0.0000									
2	3	3	2	2.5000	-4.0000	0.9000	-2.5000	-1.0000	0.0000
0.0000									
1	3	3	3	1.2329	-4.0000	1.0000	-2.5000	-1.7479	0.0000
0.0000									
2	3	3	3	0.8302	-4.0000	-0.7763	-2.5000	-1.0000	0.0000
0.0000									
3	3	3	3	-2.5000	-4.0000	1.0000	-2.5000	-1.0000	0.0000
0.0000									

5	3	3	5	1.7276	50.0000	0.0100	-2.5000	-1.0000	0.0000
0.0000									
1	1	1	5	2.1746	45.0000	0.6721	-2.5000	0.0000	0.0000
0.0000									
5	1	1	5	5.5647	24.5181	8.0000	-2.5173	0.0000	0.0000
1.1091									
2	1	1	5	0.0100	11.4523	2.1450	-3.5000	0.0000	0.0000
1.0000									
1		! Nr of hydrogen bonds;at1;at2;at3;Rhb;Dehb;vhb1							
3	2	3		2.1200	-3.5800	1.4500	19.5000		